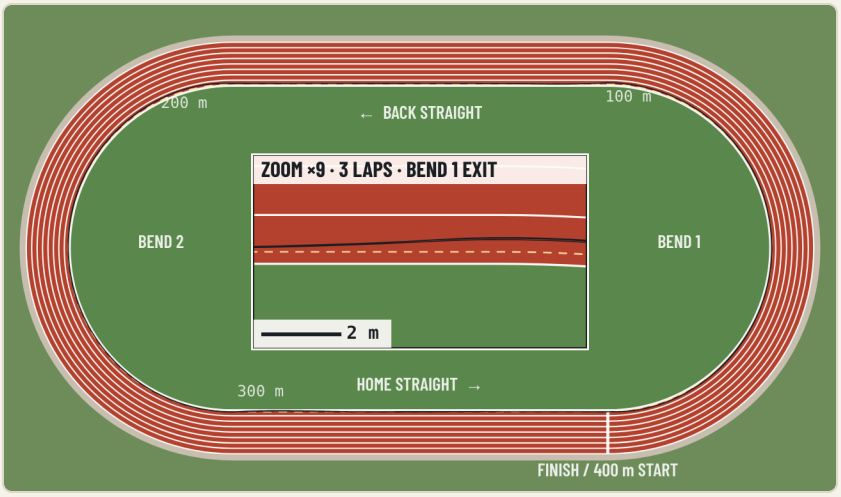
PX4 V1.17 SITL · GAZEBO HARMONIC · MAVSDK · 400 M STANDARD TRACK

A simulated drone that paces a 400 m lap to the programmed split

The track is built to the World Athletics marking plan, flown under PX4 offboard control, and timed on the autopilot's clock. Every figure is read from the flight's PX4 log: the simulator's true position, scored against a virtual runner holding the target pace.



True path of all 3 laps at 59.0 s (Gazebo x500) over the track. Dashed: lane-1 measurement line (0.30 m outside the kerb), where pace is defined. 0.17 m RMS off the line; lap-to-lap spread 0.02 m; lanes are 1.22 m wide.

±21 ms
worst of 3 laps at 59.0 s, Gazebo x500

±42 ms
worst start/finish-line crossing vs the pace clock

0.17 m
RMS gap to the virtual runner, whole race

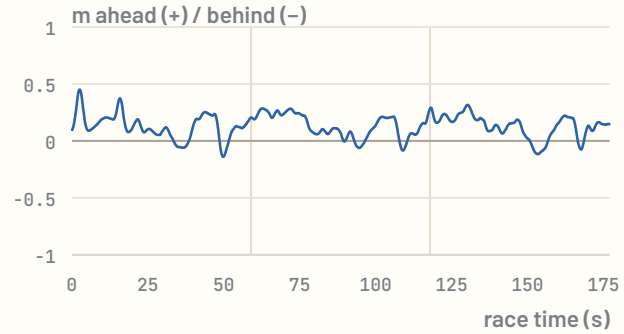
0.17 m
RMS off the lane-1 line (a lane is 1.22 m)

±61 ms
worst of 5 laps in PX4's SIH simulator, same code

Split sheet

RUN	LAP	TARGET	FLOWN	LAP Δ	LINE VS CLOCK
Gazebo	1	59.00	58.984	-16 ms	-30 ms
	2	59.00	58.988	-12 ms	-42 ms
	3	59.00	59.021	+21 ms	-22 ms
SIH	1	59.00	58.975	-25 ms	+1 ms
	2	59.00	59.001	+1 ms	+2 ms
	3	59.00	58.939	-61 ms	-59 ms
	4	59.00	59.032	+32 ms	-27 ms
	5	59.00	59.047	+47 ms	+20 ms

Gap to the virtual runner, 3 × 59.0 s (Gazebo x500)



RMS 0.17 m, worst 0.45 m. Shaded: line crossings.

How the pacer works

1 · Track
World Athletics geometry: 36.50 m kerb, 84.39 m straights, lanes 1–8; lane 1 measured 0.30 m out (400.00 m). Placed by take-off point, offset or GPS.

2 · Schedule
A virtual runner's s(t), v(t), a(t): run-up to pace before the line, per-lap targets, pace changes that still land every lap on time.

3 · Setpoints
50 Hz on PX4's clock: position, tangent velocity, turn + drag acceleration, nose on the tangent, via MAVSDK offboard.

4 · Scoring
From the ULog: line crossings at 50 Hz, gap to the runner, off-line on straights, bends and joins, height, lap-to-lap spread.

What the logs showed (findings 1–3, 5 on PX4 SIH; 4 on Gazebo)

1. **Position-only setpoints look right and run late.** Laps came in within 62 ms of 59 s, yet the drone crossed the line **1.06 s late** every lap, 7.8 m behind the runner. PX4's position loop makes the drone trail by speed ÷ MPC_XY_P (6.78 / 0.95 ≈ 7.1 m). The pacer sends position, velocity and acceleration together at 50 Hz.

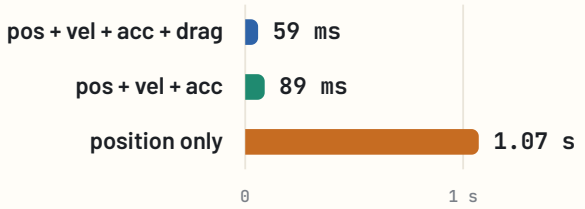
2. **A waypoint mission can't hold pace through bends.** 71 fly-through waypoints at 6.78 m/s with PX4 defaults gave laps of **98.51 s, 91.38 s** against 59 s. Bends averaged 3.53 m/s. Missions have no time target, so pacing needs the setpoint stream.

3. **Drag goes in the feed-forward, measured from a log.** Without it the drone ran 1.48 m wide on bends and lost up to 173 ms a lap. A least-squares fit on any flight log gives the coefficient: 1.008 from the SIH estimate (true 1.00), and 0.119 for the Gazebo x500 from one calibration lap.

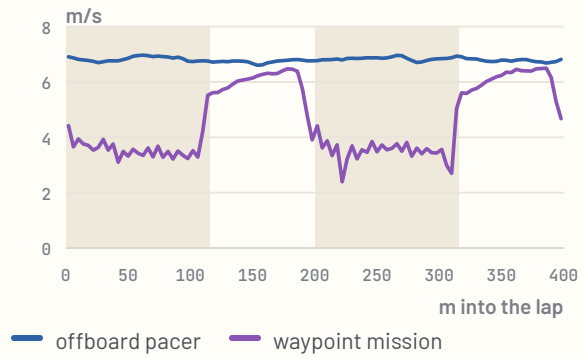
4. **A GPS-delay mismatch hides inside perfect laps.** In the first Gazebo run every lap was within 7 ms, yet every crossing was 87 ms late. PX4's default EKF2_GPS_DELAY (110 ms) against Gazebo's zero-latency GPS put the estimate 0.69 m ahead of the drone. Matched to 0, the mean crossing became –31 ms. On hardware it must match the receiver.

5. **Time comes from the autopilot.** Setpoints are computed for PX4's own clock; worst gap 54 ms against the 1000 ms offboard-loss failsafe; offboard never dropped.

Worst line crossing vs pace clock



Speed around the lap: pacer vs mission



Phase 1, and what's already working beyond it

Phase 1 scope

- PX4 SITL with Gazebo Harmonic (x500) and PX4's built-in SIH quadcopter
- Gazebo world with a standard 400 m track, 8 lanes, World Athletics dimensions
- Oval flight path: two 84.39 m straights, two 36.80 m-radius bends, continuous
- Repeatable laps: 0.02 m mean lap-to-lap spread in Gazebo
- Source, config files, setup steps for Ubuntu 22.04 / 24.04, 51 tests

Already prototyped for later phases

- Programmable lap times, per lap: 72-66-62-59-56 s flown within ±80 ms
- Constant pace through bends; flying beside a runner with corrected bend speed
- Timing and repeatability report from every flight log
- Pace planner: enter splits, check PX4 limits, export the config
- Drag identification for a new airframe from one flight

Same code in Gazebo Harmonic (x500, track400 world)

One lap with no drag term; drag identified from that lap's log (**0.119 s^{-1}**); then 3 paced laps using it, with EKF2_GPS_DELAY matched to the simulated GPS.

RUN	DRAG TERM	LAPS	WORST LAP Δ	MEAN LINE VS CLOCK	GAP RMS	OFF-LINE RMS	HEIGHT RMS
calibration lap	0.000	1	-15 ms	+91 ms	0.73 m	0.28 m	0.05 m
paced laps	0.119	3	+21 ms	-31 ms	0.17 m	0.17 m	0.03 m

Measured: real-time PX4 v1.17.0 SITL, default parameters (Gazebo: EKF2_GPS_DELAY 0), SIH quadcopter and Gazebo Harmonic 8.10 x500. Ground-truth positions and autopilot-clock times from each PX4 log.

Not measured: a real drone, wind, real GPS (non-RTK error is often 1–2 m), people near the drone. Simulator sensor noise is lighter than real hardware. Results are single runs of 1–5 laps.